Prediction models and simulation: summary

7 November 2026 · Sources: `detection/models/README.md` (model card), `models/reports/*`, `models/falldet/*.py`, `simulation/README.md`, `simulation/MODEL_API.md`, `simulation/thundersim/*`, `simulation/HOW_IT_RUNS.pdf`

> **Read first.** All training falls are lab falls by young volunteers onto mats, two of the four spine anchors in the training data are synthetic, and the simulator results are on simulated data. None of these numbers is validated real-world performance, and none supports a medical claim. On real falls of older adults, the best published feature model reached F 64.6 % at 0.56 false alarms/h (FARSEEING, Palmerini et al. 2020).

1 · The system in one paragraph

A worker wears a 4-IMU spine strip (MPU-6050s at **pelvis, lumbar, thoracic, upper thoracic**). The prediction stack, `falldet`, turns the accelerometer and gyro streams into three outputs: `normal`, `near_fall` (lost balance, then recovered) and `fall_confirmed` (alarm). A separate set of posture estimators judges bad spine posture. Because no real 4-IMU data exist yet, **ThunderSim** simulates a construction worker wearing the strip, pushes the signals through the real device and backend data path, and scores any model against 35 site scenarios, four firmware configurations and full fall-free 8 h shifts.

2 · Prediction models

MODEL	JOB	METHOD	RUNS ON	STATUS
falldet Stage 1	Fall candidate trigger	Threshold state machine on pelvis + lumbar	ESP32-portable	Shipped
falldet Stage 2	Confirm fall / near-fall / normal	Gradient-boosted trees, 28 features, 8 s aftermath window	Laptop / backend	Shipped (target + ±2 g versions)
Post-rules	Cases missing from training data	Hand-written rules after Stage 2	Laptop	Shipped, untested on real data
Near-fall v2	Stumbles, rail grabs, absorbed blows	Perturbation trigger + GBT on 22 short-window features	Laptop	Default in the stream
baseline	Reference to compare against	Research report's two-stage rules + simulator fixes	Simulator	Comparison only
Posture estimators	Bad spine posture	Accel tilt / gyro+accel fusion + ergonomic rules (EN 1005-4, ISO 11226)	Simulator	Tested in sim

Stage 1 trigger (`falldet/stage1.py`)

Per sample: `svm_g` = mean($|a_{pelvis}|$, $|a_{lumbar}|$)/g and `rot` = max($|\omega_{pelvis}|$, $|\omega_{lumbar}|$).

CONDITION	TARGET (±16 G)	CURRENT (±2 G)
Free fall	< 0.7 g, ≥ 100 ms	< 0.6 g, ≥ 100 ms
Impact	> 2.25 g	> 1.9 g
Rotation	> 3.0 rad/s	> 3.0 rad/s

Fires on impact plus at least one other condition within 1 s; 5 s refractory. Thresholds were grid-searched inside each LOSO fold as the lowest-trigger-rate setting catching ≥ 95 % of fall and near-fall recordings. About 23 triggers per hour of activity.

Stage 2 confirmer (`falldet/features.py`)

Window 2 s before to 8 s after the trigger. 28 features:

- **Multiphase** (pelvis + lumbar): free-fall dip, impact peak, jerk, peak rotation, activity before, acceleration SD after impact, did walking resume.
- **Four-anchor posture**: each anchor's tilt 2–8 s after, change from before, anchors lying (> 60°), cross-anchor spread, fraction of time still, longest still run.

`HistGradientBoostingClassifier`, depth 3, 200 iterations, balanced weights. A numpy-only logistic-regression copy (`stage2_logreg_coefficients.json`) exists for C++/JS ports (F1 0.976 vs 0.981). Tilt is estimated in-house from accel + gyro; heading is never used.

Post-rules (`models/post_rules.json`)

- **Fall from height / harness arrest:** free fall ≥ 0.35 s (≈ 0.6 m) followed by no gait → `fall_confirmed`; ≥ 0.6 s (≈ 1.8 m) always confirms. Cost: false alarms 0.73 → 0.92 /h (two VTT jump-downs).
- **Declared lying work:** a fall alarm is dropped if ≥ 3 anchors were already lying before the trigger.

Near-fall v2 (`falldet/nearfall.py`)

Built because the fall path labelled 0–5 % of simulated near-falls correctly. Trigger: trunk rotation > 1.5 rad/s plus a dip (< 0.75 g) or jolt (> 1.8 g) within 0.5 s. Decision 3 s later: all anchors back upright (< 45°) and GBT p(near_fall) ≥ 0.6. Output is a **near-miss log entry, never an alarm**.

Training data

DATASET	CONTENT	LICENCE
Erciyes / UCI 455	17 volunteers, 20 fall + 16 ADL types, waist + chest IMUs at 25 Hz, 3,303 recordings	CC BY 4.0
VTT-ConIot	13 construction workers × 16 activities, hip + back IMUs at 100 Hz, 3.3 h	CC BY 4.0

Pelvis and upper-thoracic anchors are **real**; lumbar and thoracic are **synthesised** by slerp between them (f = 0.30, 0.65) with rigid-trunk acceleration. Harmonisation converts units, calibrates sensor-to-segment from the data, resamples to 100 Hz, fuses orientation with a Mahony filter, then emulates the MPU-6050 per firmware profile (bias, noise, DLPF, clipping, int16). Finding: the Erciyes chest and thigh sensor IDs are swapped in its documentation.

Lab results (leave-one-subject-out, 30 folds)

FIRMWARE / MODEL	SENSITIVITY	SPECIFICITY	F1	3-CLASS MACRO-F1	FALSE ALARMS/H
Target (±16 g, 44 Hz), GBT, shipped	0.969	0.995	0.981	0.951	0.73 (0.92 with post-rules)
Target, logistic regression	0.959	0.995	0.976	0.945	0.73
Target, report's hand rule	0.764	0.997	0.864	0.816	0.46
Current (±2 g, 10 Hz), retrained GBT	0.966	0.995	0.980	0.952	0.64
Current data, target-trained model	**0.738**	0.999	0.849	0.798	0.09

- **Weak spots:** syncope sliding down a wall (74 % confirmed); stumble with recovery (fall path flagged 39 %; near-fall v2 catches 59 % of real lab stumbles). Lab false alarms come from lying down on a bed and sitting hard on a sofa. Construction (VTT): 0 false alarms in 3.3 h.
- **Four anchors don't help yet:** pelvis-only features score F1 0.982 vs 0.981 for all four. Expected, since two anchors are synthetic; it needs real 4-IMU recordings to test.
- **Using the wrong model for the firmware costs ~23 points of sensitivity** because ±2 g clips the impact.

3 · Simulation strategy: ThunderSim

Goal: test fall and man-down models end to end before anyone wears the strip: thresholds, timers, firmware settings, packet loss, dead sensors, false alarms over a whole shift. It is pure Python (numpy, scipy, protobuf, matplotlib), not a game or physics engine, because the aim is to reproduce what four MPU-6050s and the data path would report, not to animate a body.

Pipeline

Scenario + seed → 4-link spine body @ 1 kHz → Fall + site physics → Soft tissue / strap → MPU-6050 emulator → ImuStripData protobuf → BLE link → Real backend decoder → Neutral calibration → Model → 15 s worker check-in → Score vs ground truth

Body and motion (`spine.py`, `motions.py`)

- Four rigid links (pelvis → upper thoracic), soft range-of-motion limits, trunk speed capped at 180°/s.
- Normal movement is scripted minimum-jerk kinematics, C1-continuous so no velocity step looks like a fake impact.
- Breathing, sway and fidget always on (an unconscious worker still breathes; a strip on a table does not).
- Sensors sit 4 cm behind the spine, so rotation adds lever-arm terms $\omega\times(\omega\times r)$ and $\alpha\times r$.
- Soft-tissue/strap mount filter per anchor, 8–25 Hz, random mounting tilt ±6° (±9° in winter clothing).

Fall physics (`falls.py`), only during the fall

- Loss of balance: inverted pendulum with knee collapse; slips by friction.
- Flight: ballistic, free-fall time = $\sqrt{(2h/g)}$ (tested).
- Body-first impact: 0.35 × body mass, hip stiffness 39–55 kN/m in series with the surface; hip speeds 3.3–4.0 m/s vs consensus 3.4 ± 0.5.
- Harness arrest within OSHA limits (≤ 1.8 m free fall, ≤ 8 kN), then hanging 15–35° forward.
- Stairs, roof and excavation slides, fainting collapse, struck-by, near-falls.

Environment (`site.py`, `device.py`, `transport.py`)

- **Worker:** 1.55–1.95 m, 55–120 kg, 3–15 kg of tools and PPE.
- **Surfaces:** concrete, steel/plywood deck, scaffold plank, rebar mat, gravel, soil, mud, crash mat, safety net; each with stiffness, damping and friction.
- **Conditions:** dry, dusty, wet, muddy, oily, icy; slip when $v^2/(2\mu g) > 0.15$ m.
- **Device:** MPU-6050 register-map values: DLPF, noise, bias, temperature-driven gyro drift, int16, clipping, sequential mux reads 0.8–1.5 ms apart.
- **Radio:** bursty Gilbert–Elliott packet loss, 50–300 ms outages, 7.5–60 ms latency, throughput cap.
- **Data path:** the real `backend/app/protocol.py` decoder, imported unchanged.

Every parameter is tagged **[src]** sourced, **[design]** randomised starting value, or **[verify]** sourced but unconfirmed.

What gets varied

SCENARIO GROUP (35 TOTAL)	EXAMPLES	CORRECT OUTPUT
Falls from height	ladder, scaffold, roof, floor opening, trench (1–6 m)	`fall_confirmed`
Same-level falls	slip, trip, stairs, struck-by knock-down	`fall_confirmed`
Arrested falls	harness, safety net	`fall_confirmed`
Collapse	syncope standing / kneeling	reported only
Near-falls	stumble, rail grab, absorbed blow, friction-dependent slip	`near_fall`
Construction hard negatives	hammering, jackhammer, equipment cab, kneeling, jump-down, hard sit, lying work, lunch nap, hard hat knocked off, strap slip, strip on a table…	`normal`

FIRMWARE PROFILE	RATE	ACCEL	GYRO	DLPF

`current` (as flashed)	0.5 Hz	±2 g	±250 °/s	10 Hz
`current_100hz` (period fixed)	100 Hz	±2 g	±250 °/s	10 Hz
`target` (recommended)	100 Hz	±16 g	±2000 °/s	44 Hz
`target_8g`	100 Hz	±8 g	±1000 °/s	44 Hz

Fall-free shifts chain 1–10 min activity blocks in SIM_SPEC 5.2 proportions for 8 h, plus about 10 jump-downs and 5 runs, to measure false alarms per shift. **Honest-testing rules:** tune only on dev seeds 0–19, report once on held-out seeds ≥ 1,000,000, Wilson 95 % intervals, re-run with one anchor dead.

Model contract (`MODEL_API.md`)

Any class with `reset(context)`, `update(batch)` and optional `finish()` plugs in via `--models package.module:Class`. Batches are ~0.5 s of `t`, `acc` (N,4,3) m/s², `gyro` (N,4,3) rad/s, dead-slot and clipping masks, and `work_mode`. Models must be causal. Every alert opens a 15 s worker check-in; only uncancelled alerts reach a supervisor.

4 · What the simulation found

FALLDET, TARGET FIRMWARE	DEV SEEDS	TEST SEEDS	PASS BAR
Falls from height confirmed	98 %	100 %	≥ 90 %
Same-level, stairs, struck-by confirmed	95 %	95 %	≥ 80 %
Harness arrest confirmed	100 %	100 %	≥ 80 %
Alert latency, 95th percentile	8.3 s	8.3 s	≤ 30 s
False alarms per 8 h shift (model alone)	**7.8**	–	≤ 1

- **Firmware as flashed (0.5 Hz) detects 0 of 280 falls.** Fixing only the sampling period recovers almost all of them; ±2 g and the 10 Hz filter matter much less, except for slow syncope (65 % → 10 %).
- **False alarms** come from hard sits, equipment rides and jump-downs followed by standing still. The 15 s check-in cancelled every one; none reached a supervisor (latency with check-in 23.3 s).
- **Near-falls:** 0 % with the v1 path; near-fall v2 scores 95–100 % on simulator near-fall scenarios with 0 flags per shift, but only 59 % on real lab stumbles.
- **One dead anchor:** falldet keeps 97–98 % sensitivity without the pelvis or the upper-thoracic sensor.
- **Calibration gap:** simulated mat falls peak at 6.5–7 g against 3.95 g median for Erciyes mat falls. Volunteers fell in a controlled way; site falls were not scaled down, so a lab-trained model will see larger impacts on site.
- **Open item:** the newly agreed device spec delivers ~20 Hz over BLE. As shipped, falldet's 40 Hz floor catches 0 of 12 falls at 20 Hz; with the floor at 15 Hz it matches 100 Hz (10 of 12). Needs a new profile, a re-run and agreement with research-95.

5 · 3D viewer (new, viewer only)

`python -m thundersim viz` serves a local three.js page (vendored, no internet) at `http://127.0.0.1:8765`. It runs a scenario through the same chain as `run` and replays it with a scene matching the scenario (ladder, scaffold edge, harness and lanyard, safety net, stairs, equipment cab…). The torso follows the simulated spine exactly; legs, arms and head are **illustrative**, posed by a procedural rig (`viz/rig.py`) with planted feet, footsteps timed to simulated heel strikes and two-bone IK. Sensor pods show the true chip orientation, and arrows show what each IMU actually reported after clipping, sample-and-hold and packet loss. A `motionlib` with a licence guard is being set up to retarget openly licensed mocap onto the limbs. None of this changes the simulated signals or the detector.

6 · Limits and next steps

- **Data:** lab mat falls by young volunteers; Erciyes is 25 Hz (true bandwidth 12.5 Hz), so impact thresholds will need re-tuning on real 100 Hz recordings. Two anchors synthetic.

- **Simulator:** kinematic body with physics only during falls, single spring-damper contact, many [design] values not yet checked against real hardware. A pass is necessary, not sufficient.
- **Alarm rate:** ~0.9 false confirmations/h (6–7 per shift) is too many for a live alarm without the worker check-in.
- **Untested rules:** harness-arrest and lying-work post-rules are hand-written, not learned.
- **Next:** fix firmware sampling (≥ 100 Hz acquisition, raw accel delivered); add the agreed 20 Hz profile and re-run; record the team's own MPU-6050s and strap to replace [design] values; collect real 4-IMU mat falls, daily activity and shadow-mode shifts to test whether four anchors beat one.

Key commands

WHERE	COMMAND	WHAT
`detection/models`	`.venv/bin/python run_demo.py [--profile current]`	Replay fixtures through the detector
`detection/simulation`	`python -m thundersim run H-LAD --seed 3 --model falldet --profile target`	One scenario run with plot
	`python -m thundersim suite --models baseline,falldet --profiles target,current --runs 20`	Randomised suite + report
	`python -m thundersim shifts --models falldet --count 10 --hours 8`	Fall-free shift false alarms
	`python -m thundersim posture --runs 10` · `viz`	Posture test · 3D viewer